

Technical Datasheet — T8-40g

VBF-T8R1-DSH-01 Case t8_r1 — Long-endurance drone battery 2026-08-25

Prepared: Reviewed: Approved: _____

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Field	Value
Rated capacity (Ah)	Nominal 6.1494 (parameter set, scaled); verified 6.9427 (1C DFN simulation)
Nominal voltage / voltage window (V)	Window 2.5 – 4.2; discharge midpoint 3.876 (parameter set / simulation)
Rated energy (Wh)	25.246 (time integration of V·I, 1C discharge)
Energy density (Wh/kg)	634.32 (contract-caliber mass formula; electrolyte excluded — annotated)
Energy density (Wh/L)	1027.1 (stack volume caliber)
Maximum continuous discharge rate	5C verified: 6.8571 Ah, retention 98.77%, T_max 319.99 K (DFN, 25 °C ambient)
Fast-charge capability	4C @45 °C: T_max 330.41 K (≤ 333.15 K), no plating (anode min +0.0082 V); acceptance voltage-limited to 0.94 Ah — 4C for top-up only, ≤2C for full charge
Operating temperature range	25 °C discharge / 45 °C charge simulated; other temperatures not simulated (Not provided)
Cycle life	Not simulated (requires aging model) — not fabricated
Safety determination	4C-45 °C temperature rise PASS; plating PASS (protocol defaults; see DVPR)
DC internal resistance / power density	0.684 mΩ / 156.1 kW/kg (mechanical derivation)
Dimensions	72.09 × 1752.2 × 0.195 mm (unwound electrode geometry; shell thickness not provided)
Mass (g)	39.800 (contract caliber, electrolyte excluded; +11.03 g electrolyte at lit. density 1.2 g/cm³)